

화학수학

# 스무 번째 수업

시험 안내 및 복습



# 기말고사 안내

- 시간: 12월 5일(목) 오후 3시~4시
- 장소: 본인의 학번 마지막 자리가
  - 1, 2, 3, 4인 경우: 화학관 B101호
  - 5, 6, 7, 8인 경우: 화학관 112호
  - 9, 0인 경우: 화학관 409호
- 계산기 불필요, 신분증 지참
- 10분 전까지 입실 완료

# 시험 범위

미분방정식(12장)

연산자, 행렬, 군론(13장 + 14장 일부)

확률(15장)

수업 시간에 다루지 않은 내용은 범위에 포함되지 않음

# 12. Differential Equations

## (12장 미분방정식)

# Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

# Terminology

- Order of DE (계수): order of the highest derivative
- Linear (선형) vs. nonlinear (비선형)
  - Linear: contains the function ( $y$ ) and its derivatives only to the first power (Note: terms in  $x$  can be nonlinear)

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Nonlinear: otherwise

$$(1 - y) \frac{dy}{dx} + 2y = e^x, \quad \frac{d^2 y}{dx^2} + \sin y = 0, \quad \dots$$



# Linear Differential Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Homogeneous (동차):  $g(x) = 0$
- Inhomogeneous (비동차):  $g(x) \neq 0$

# General and Particular Solutions

## (일반해와 특수해)

- **Solution:** any function that reduces the DE to an identity when substituted into the DE

For a DE of order  $n$ ,

- **General solution:** contains  $n$  arbitrary constants
- **Particular solution:** contains no arbitrary constants
  - Particular solutions can be generated from the general solution





## Additional Conditions

- For a DE of order  $n$ , if  $n$  additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first  $(n - 1)$  derivatives at a single point
- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

# Solving a Differential Equation

1. Obtain the **general solution**: hardest part!
  2. If initial/boundary conditions are given, substitute the conditions to **determine the coefficients**
- No universal method to solve all DEs
  - For specific forms of DEs, we have general solving methods

# First-Order ODEs

$$F\left(y, \frac{dy}{dx}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

# Separable Variables

- Separable DE:

$$g(y) \frac{dy}{dx} = f(x)$$

This DE can be manipulated into

$$g(y) dy = f(x) dx$$

$$\int g(y) dy = \int f(x) dx + C$$

This equation provides the **general solution**



# Exact Differential Equations

- Pfaffian form:

$$M(x, y) \, dx + N(x, y) \, dy = 0$$

If the differential equation is **exact**,  
there is a function  $f(x, y)$  such that

$$df = M(x, y) \, dx + N(x, y) \, dy = 0$$

which implies that  $f(x, y) = k$  (constant)

# Linear Equations

$$a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{dy}{dx} + P(x)y = f(x)$$

- General solution:

$$y = e^{-\int P(x)dx} \int e^{\int P(x)dx} f(x) dx + C e^{-\int P(x)dx}$$

# Higher-Order ODEs: Linear Case

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

- Superposition principle (§ 12.2.1)
- Reduction of order (not in Mortimer)
- Linear ODEs with constant coefficients (§ 12.2, 12.3)
  - Homogeneous (§ 12.2)
  - Inhomogeneous (§ 12.3)
- We will not discuss nonlinear ODEs



# Homogeneous Linear Equations

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

- Superposition principle (중첩 원리):

If  $y_1(x), y_2(x), \dots, y_k(x)$  are the solutions of the ODE, the linear combination

$$y = c_1 y_1(x) + c_2 y_2(x) + \cdots + c_k y_k(x)$$

is also a solution of the ODE for arbitrary constants  $c_1, c_2, \dots, c_k$ .





# Reduction of Order

$$a_2(x) \frac{d^2 y}{dx^2} + a_1(x) \frac{dy}{dx} + a_0(x)y = 0$$

If a solution  $y_1$  is known,  
we can obtain the other solution  $y_2$

$$y_2(x) = y_1 \int \frac{1}{y_1^2} e^{-\int P dx} dx$$

## Homogeneous Linear ODEs with Const. Coeff.

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

- Trial solution (시행해):  $y = e^{\lambda x}$

$$a\lambda^2 e^{\lambda x} + b\lambda e^{\lambda x} + ce^{\lambda x} = 0$$

- Characteristic equation (고유방정식)

$$a\lambda^2 + b\lambda + c = 0$$

# Roots of Characteristic Equation

$$a\lambda^2 + b\lambda + c = 0$$

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

1. Two roots are real and distinct ( $b^2 - 4ac > 0$ )
2. Two roots are real and equal ( $b^2 - 4ac = 0$ )
3. Two roots are complex numbers ( $b^2 - 4ac < 0$ )

## (1) Distinct Real Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad \lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

We find two solutions:  $y_1 = e^{\lambda_1 x}$  and  $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

## (2) Repeated Real Roots

$$\lambda_1 = \lambda_2 = \frac{-b}{2a}$$

We obtain one solution:  $y_1 = e^{\lambda_1 x}$

Using the reduction of order,

$$y_2 = y_1 \int \frac{e^{-\int P dx}}{y_1^2} dx = e^{\lambda_1 x} \int \frac{e^{2\lambda_1 x}}{e^{2\lambda_1 x}} dx = x e^{\lambda_1 x}$$

$$y = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x}$$

### (3) Conjugate Complex Roots

$$\lambda_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} = \alpha + i\beta$$

$$\lambda_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a} = \alpha - i\beta$$

We find two solutions:  $y_1 = e^{\lambda_1 x}$  and  $y_2 = e^{\lambda_2 x}$

The general solution is

$$y = e^{\alpha x} (c'_1 \cos \beta x + c'_2 \sin \beta x)$$

# Inhomogeneous Linear ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = \underline{g(x)}$$

inhomogeneous term  
(비동차항)

- Complementary equation (보상방정식)

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = 0$$

## Solving Inhomogeneous ODE

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

1. Solve the complementary equation  $\rightarrow$   
complementary function (보상함수)  $y_c$
2. Find a particular solution  $y_p$  that satisfies the DE
3. The general solution is  $y = y_c + y_p$



# Trial Solutions for $y_p$

| Inhomogeneous term           | Trial solution                                           |
|------------------------------|----------------------------------------------------------|
| 1                            | $A$                                                      |
| $t^n$                        | $A_0 + A_1 t + A_2 t^2 + \dots + A_n t^n$                |
| $e^{\alpha t}$               | $A e^{\alpha t}$                                         |
| $t^n e^{\alpha t}$           | $e^{\alpha t} (A_0 + A_1 t + A_2 t^2 + \dots + A_n t^n)$ |
| $e^{\alpha t} \sin(\beta t)$ | $e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$       |
| $e^{\alpha t} \cos(\beta t)$ | $e^{\alpha t} [A \cos(\beta t) + B \sin(\beta t)]$       |

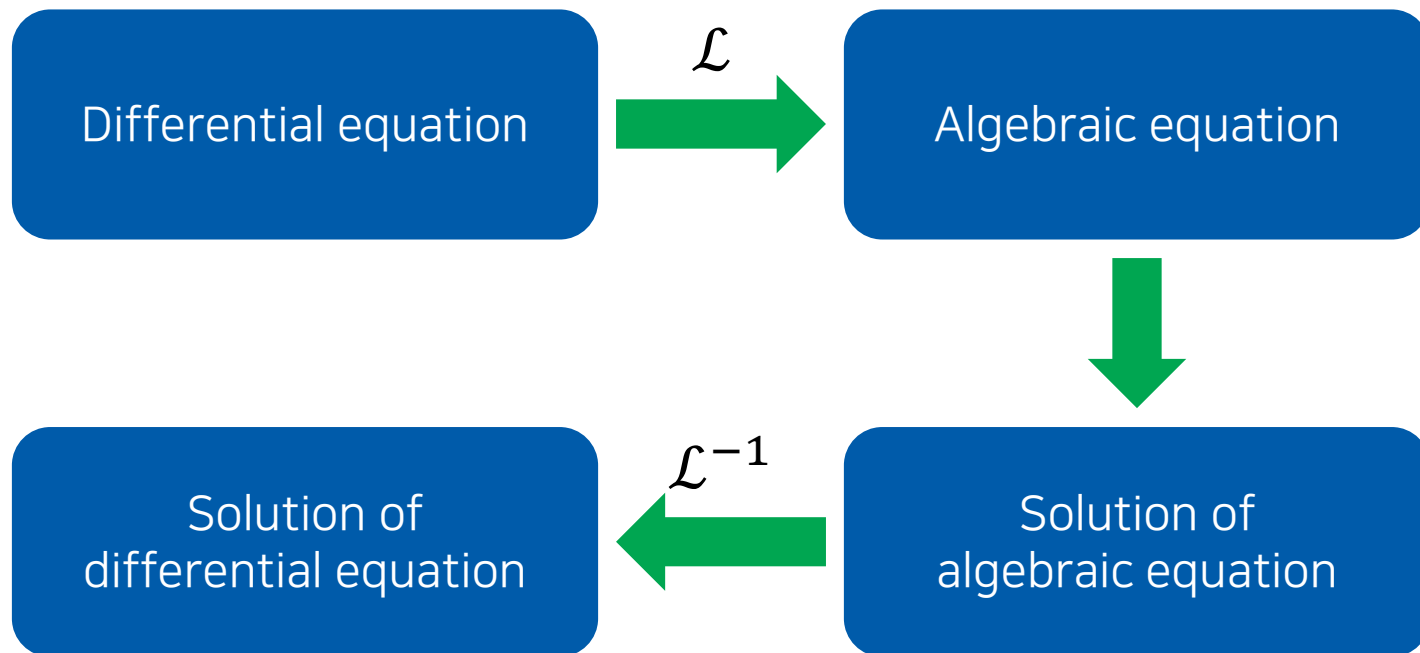
# Glitch in the Method

A trial solution  $y_p$  can be  
a solution of the complementary equation.

In this case, try  $x^n y_p$  instead!

( $n$ : smallest integer that resolves the problem)

# Solving DE with Laplace Transforms



# Partial Differential Equations

- General form of a linear second-order PDE

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

where  $A, B, \dots, G$  are functions of  $x$  and  $y$ .

- $G(x, y) = 0$ : homogeneous
- $G(x, y) \neq 0$ : inhomogeneous

## Separation of Variables

$$A \frac{\partial^2 u}{\partial x^2} + B \frac{\partial^2 u}{\partial x \partial y} + C \frac{\partial^2 u}{\partial y^2} + D \frac{\partial u}{\partial x} + E \frac{\partial u}{\partial y} + Fu = G$$

Assume that  $u(x, y) = X(x)Y(y)$ .

$$\begin{aligned} X' &= dX/dx, X'' = d^2X/dx^2, \dots \\ Y' &= dY/dy, Y'' = d^2Y/dy^2, \dots \end{aligned}$$

If we obtain  $f(x, X, X', \dots) = g(y, Y, Y', \dots)$ , let

$$f(x, X, X', \dots) = g(y, Y, Y', \dots) = \lambda$$

where  $\lambda$  is the **separation constant**.

(Tip: it is often convenient to use  $\lambda = \kappa^2$  or  $-\kappa^2$ )

# 13. Operators, Matrices, and Group Theory

(13장 연산자, 행렬, 군론)

## Mathematical Operators (연산자)

perform **mathematical operations** on some function

*e.g.* "+3" operator, "×2" operator, "d/dx" operator

- Typically denoted by a letter with a **caret** (^)
- Operate on the function on its right

## Special Operators

- Identity operator (항등 연산자)  $\hat{E}$ : multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자)  $\hat{0}$ : multiplication by 0

$$\hat{0}f = 0$$



# Operator Algebra

- Sum of two operators: if  $\hat{C} = \hat{A} + \hat{B}$ , then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if  $\hat{C} = \hat{A}\hat{B}$ , then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

# Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A} ?$$

## More on Commutative Property

- If  $\hat{A}\hat{B} = \hat{B}\hat{A}$ ,  $\hat{A}$  and  $\hat{B}$  are said to **commute**
- **Commutator** (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If  $\hat{A}$  and  $\hat{B}$  commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

# Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A} \cdots \hat{A} \text{ (} n \text{ times)}$$

## Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$ : inverse of operator  $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$  is also the inverse of  $\hat{A}^{-1}$
- Not all operators have inverses (*e.g.*  $\hat{0}$ )

# Eigenfunctions and Eigenvalues

## (고유함수와 고윳값)

For a given operator  $\hat{A}$ , a function  $f$  such that

$$\hat{A}f = af$$

is called an **eigenfunction** of  $\hat{A}$ , and  $a$  is called an **eigenvalue**. This equation is called an **eigenvalue equation** (고윳값 방정식).

## Hermitian Property (에르미트 성질)

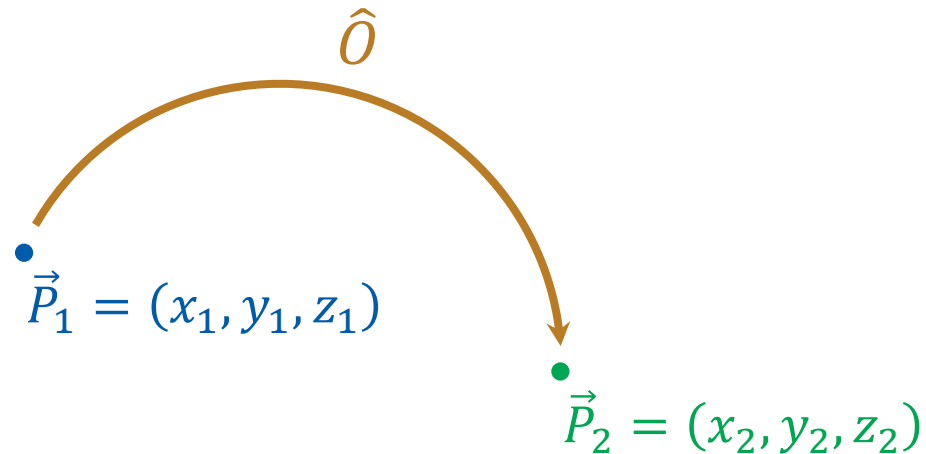
- If  $\hat{A}$  is **Hermitian**, it obeys the relation

$$\int \chi^* (\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

where  $\chi$  and  $\psi$  are arbitrary functions and  $q$  represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

# Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$



# Point Symmetry Operators

## (점대칭 연산자)

a class of symmetry operators  
that leave **at least one point unmoved**  
(the unmoved point = origin)

Identity operator  $\hat{E}$

Inversion operator  $\hat{i}$

Reflection operator  $\hat{\sigma}$

Rotation operator  $\hat{C}_n, \hat{S}_n$

## Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, **the symmetry operator belongs to the object.**
- Coordinate system
  - origin: center of mass of the object
  - z axis: the rotation axis of highest order (largest  $n$ )

# Symmetry Operators on Functions

If an operator  $\hat{O}$  operates as  $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$ ,  
for a function  $f$ ,

$$\hat{O}f = g$$

where  $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$ .

Use the relation between  $(x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$ !

# Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

the operator leaves a function unchanged

- Eigenvalue = -1

the operator changes the sign of the function's value at all points

# Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

matrix elements  
(행렬 원소)

( $m$  by  $n$  matrix)

# Matrix Algebra

- Two matrices are **equal** if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix

- **Sum of two matrices:**  $\mathbf{C} = \mathbf{A} + \mathbf{B}$  if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- **Product of a scalar  $c$  and a matrix  $\mathbf{A}$ :**  $\mathbf{B} = c\mathbf{A}$  iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

## Product of Two Matrices

- Matrix product  $\mathbf{C} = \mathbf{AB}$  iff

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

where  $n$  is the number of columns in  $\mathbf{A}$  and the number of rows in  $\mathbf{B}$ .

$$(m \text{ by } n \text{ matrix}) \times (n \text{ by } p \text{ matrix}) = (m \text{ by } p \text{ matrix})$$

# Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$



# Identity Matrix (항등행렬) **E**

$$\mathbf{EA} = \mathbf{AE} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$E_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

# Zero Matrix (영행렬) $\mathbf{0}$

$$\mathbf{0A} = \mathbf{A0} = \mathbf{0}$$

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$0_{ij} = 0$$

## Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^n a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

# Gauss-Jordan Elimination

- Construct an **augmented matrix** (첨가행렬)

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[ \begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[ \begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array} \right]$$

- Then make the identity matrix

$$\left[ \begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array} \right] A^{-1}$$

# Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows  $\rightarrow$  columns, columns  $\rightarrow$  rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$(\tilde{\mathbf{A}})_{ij} = \tilde{a}_{ij} = a_{ji}$$

# Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

(에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$(\mathbf{A}^\dagger)_{ij} = a_{ji}^*$$

## Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse
- Symmetric matrix (대칭행렬):  $\mathbf{A} = \tilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬):  $\mathbf{A} = \mathbf{A}^\dagger$
- Orthogonal matrix (직교행렬):  $\mathbf{A}^{-1} = \tilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬):  $\mathbf{A}^{-1} = \mathbf{A}^\dagger$

## Trace of a Square Matrix (대각합)

$$\text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \text{tr } \mathbf{A} = a_{11} + a_{22} + a_{33}$$

Even when  $\mathbf{AB} \neq \mathbf{BA}$ ,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$



# Determinant of a Square Matrix (행렬식)

$$\det \mathbf{A} = \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix}$$

- 2 by 2 determinant

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{12}a_{21}$$

## Expanding the Determinant

1. Pick a row or a column (with many zeros)
2. Calculate  $\text{each element} \times \text{sign} \times \text{minor}$ 
  - **Minor**: determinant after deleting the row and the column containing the element
  - **Sign**: counting the number of steps of one row or one column required to get from the upper left element  
→ even number: (+), odd number: (-)
3. Sum them up across the row or the column
4. Repeat this up to a sum of 2 by 2 determinants



# Applications of the Determinant

- Slater determinant
- Vector product of two vectors
- Determinant of a singular matrix
- Matrix eigenvalue problem

# Matrix Eigenvalue Problem

For a given square matrix  $\mathbf{B}$ , if we have

$$\mathbf{B}\vec{x} = b\vec{x}$$

- $\vec{x}$  (column vector): eigenvector
- $b$  (scalar): eigenvalue

$$(\mathbf{B} - b\mathbf{E})\vec{x} = \vec{0}$$

$$\det(\mathbf{B} - b\mathbf{E}) = 0$$

# Group (군)

Collection of members  
with specific mathematical properties

## Requirements for a Group

1. If  $A$  and  $B$  are members of the group, then the **product**  $AB$  is also a member of the group
2. The group must contain the **identity**  $E$  such that

$$AE = EA = A \text{ (for every member } A)$$

3. The **inverse** of every member  $A^{-1}$  is a member

$$A^{-1}A = AA^{-1} = E$$

4. The **associative law** holds:  $A(BC) = (AB)C$

## Point Group (점군)

Formed by a set of point symmetry operators  
belonging to a symmetrical object

# Representation of a Point Group

Symmetry operator  $\leftrightarrow$  Matrix

$$\hat{A}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\mathbf{A} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \begin{bmatrix} x_2 \\ y_2 \\ z_2 \end{bmatrix}$$



# 15. Probability, Statistics, and Experimental Errors

(15장 확률, 통계, 실험오차)

## Probability Distribution (확률분포)

mathematical function that gives  
the probabilities of occurrence of  
different possible outcomes for an experiment

- Discrete probability distribution (이산 확률분포):  
possible outcomes have discrete values
- Continuous probability distribution (연속 확률분포):  
possible outcomes have continuous values

# Probability Distributions

|                                      | Discrete                                      | Continuous                                                    |
|--------------------------------------|-----------------------------------------------|---------------------------------------------------------------|
| Variable                             | $x_1, x_2, \dots, x_n$                        | $x$ : real in $[x_{\min}, x_{\max}]$                          |
| Probability distribution             | $p_i$                                         | $f(x) dx$                                                     |
| Normalization                        | $\sum_{i=1}^n p_i = 1$                        | $\int_{x_{\min}}^{x_{\max}} f(x) dx = 1$                      |
| Mean value<br>(normalized dist.)     | $\langle x \rangle = \sum_{i=1}^n x_i p_i$    | $\langle x \rangle = \int_{x_{\min}}^{x_{\max}} x f(x) dx$    |
| Expected value<br>(normalized dist.) | $\langle g \rangle = \sum_{i=1}^n g(x_i) p_i$ | $\langle g \rangle = \int_{x_{\min}}^{x_{\max}} g(x) f(x) dx$ |

# Examples

- Discrete probability distribution
  - Binomial probability distribution
- Continuous probability distribution
  - Uniform probability distribution
  - Gaussian probability distribution

# Application: Quantum Mechanics

- For a wavefunction  $\psi$  of the system,

Probability distribution  $\propto |\psi|^2$

- Expectation value of an observable  $A$

$$\langle A \rangle = \int \psi^* \hat{A} \psi \, dq$$

where  $\hat{A}$  is the operator corresponding to the variable  $A$

## Application: Kinetic Theory of a Gas

- **Boltzmann factor:** in the thermal equilibrium of temperature  $T$ , the probability of having energy  $E$  is

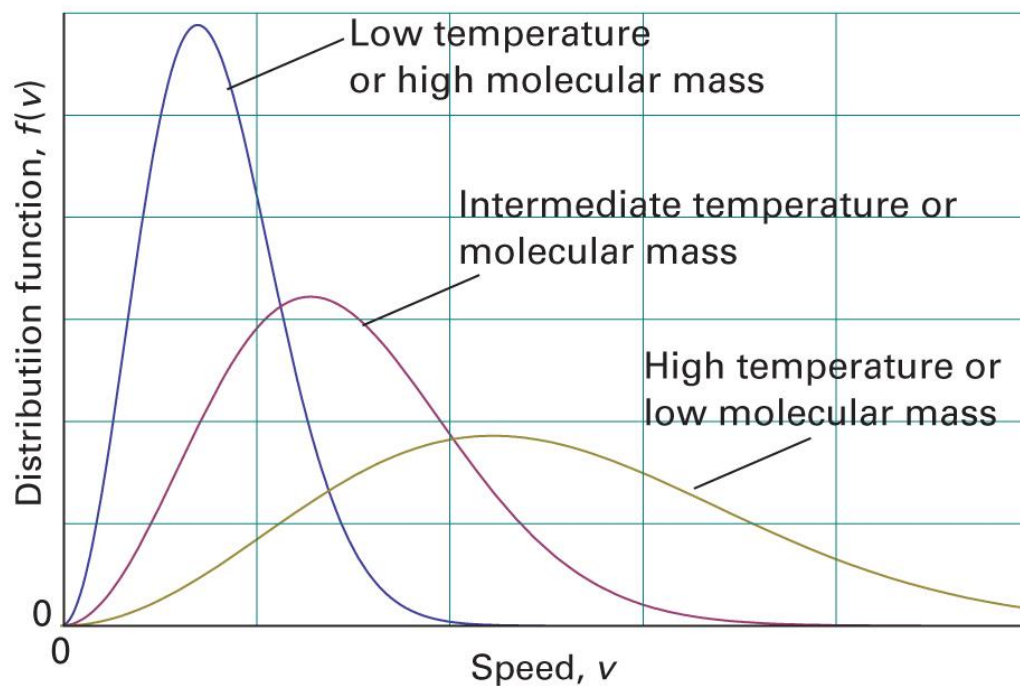
$$f(E) \propto e^{-E/k_B T}$$

- The probability that a gas molecule has the velocity  $(v_x, v_y, v_z)$  is

$$f(v_x, v_y, v_z) = \left( \frac{m}{2\pi k_B T} \right)^{3/2} \exp \left( -\frac{m(v_x^2 + v_y^2 + v_z^2)}{2k_B T} \right)$$

# Maxwell-Boltzmann Distribution

$$f_v(v) = 4\pi \left( \frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-mv^2/2k_B T}$$



(Image: Atkins *et al.*, *Atkins' Physical Chemistry*, 11<sup>th</sup> ed., Figure 1B.4)